

Quantum Data Compression



- **Aim:** Find an encoding protocol that maps the states of a Hilbert space into a new Hilbert space of reduced dimensionality, with the possibility of recovering the original states.
- Let $d = \dim \mathcal{H}$ and its L -extension $\mathcal{H}^{\otimes L}$ of dimension d^L where the source state sequences (quantum sourcewords) are defined.
- Map these sequences into quantum codewords in a new Hilbert space $\mathcal{H}_c$ s.t.
 $d_c := \dim \mathcal{H}_c < \dim \mathcal{H}^{\otimes L} = d^L$

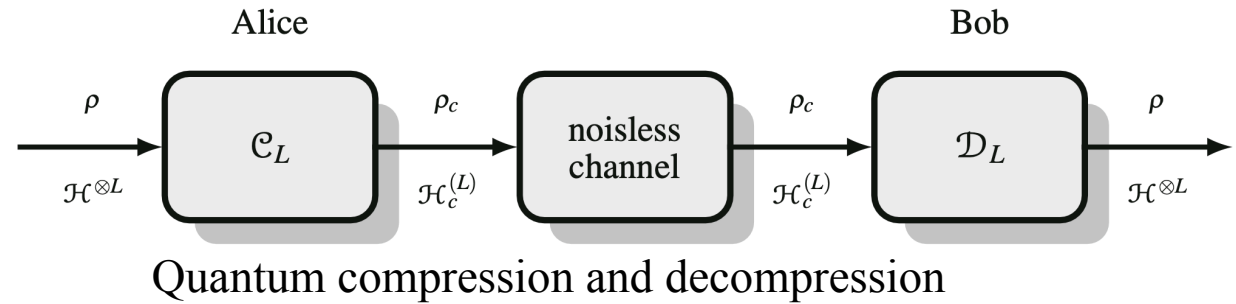


Schumacher's Compression Protocol

- Quantum compression and decompression:
 $\mathcal{C}_L : \mathcal{H}^{\otimes L} \rightarrow \mathcal{H}_c^{(L)}, \quad \mathcal{D}_L : \mathcal{H}_c^{(L)} \rightarrow \mathcal{H}^{\otimes L}$

- Compression Rate:

$$R := \frac{\log \dim \mathcal{H}_c^{(L)}}{\log \dim \mathcal{H}^{\otimes L}} = \frac{\log \dim \mathcal{H}_c^{(L)}}{L \log d}$$



- If $\mathcal{H}^{\otimes L}$ consists of L qubits ($d = 2$), then $R = \log \dim \mathcal{H}_c^{(L)} / L$, and $\dim \mathcal{H}_c^{(L)} = 2^{LR}$, thus R reads in qubits/symbol.
- $\mathcal{C}_L$ compresses ρ of $L \log d$ qubits into state ρ_c of LR qubits and $\mathcal{D}_L$ decompresses ρ_c into ρ .

- Fidelity:** $F(\rho, \sigma) = \left(\text{tr} \sqrt{\sqrt{\rho} \sigma \sqrt{\rho}} \right)^2$

s.t. $0 \leq F \leq 1$ and becomes $F = 1$ only when the two states coincide.

- Theorem:** (Schumacher's noiseless quantum channel coding) *Consider an IID quantum source with entropy $S(\rho)$ and rate R . Then, if $R > S(\rho)$ there exists an asymptotically reliable compression protocol (where $F \rightarrow 1$ as $L \rightarrow \infty$). Conversely, if $R < S(\rho)$ any compression protocol is not reliable.*

Example: Schumacher's Compression Protocol



- Consider a qubit source emitting: $|0\rangle$ w.p. 0.9 and $|1\rangle$ w.p. 0.1
- Density matrix: $\rho = 0.9|0\rangle\langle 0| + 0.1|1\rangle\langle 1| = \begin{bmatrix} 0.9 & 0 \\ 0 & 0.1 \end{bmatrix}$ with eigenvalues $\lambda_1 = 0.9, \lambda_2 = 0.1$
- Von-Neumann Entropy: $S(\rho) = -0.9 \log_2(0.9) - 0.1 \log_2(0.1) \approx 0.469$
- Interpretation: Suppose the source emits **1000 qubits**.
 - Without compression: 1000 qubits
 - With Schumacher's Compression: $\approx 1000 \times 0.469 = 469$ qubits
 - So compression is possible to 469 qubits while still recovering the original quantum information with arbitrarily high fidelity for large blocks.

Overview



- Motivation
- Quantum Entropy
- Quantum Data Compression
- **Quantum Channels**
- Quantum Capacity

Quantum Channels



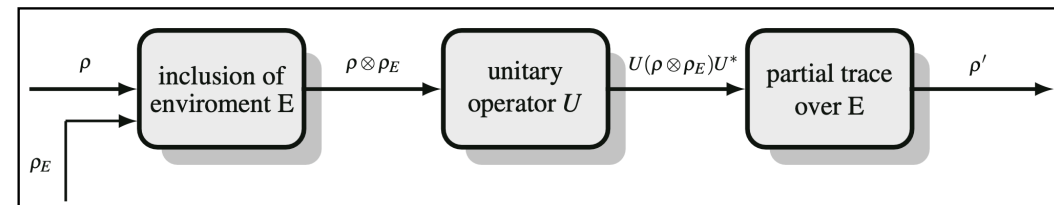
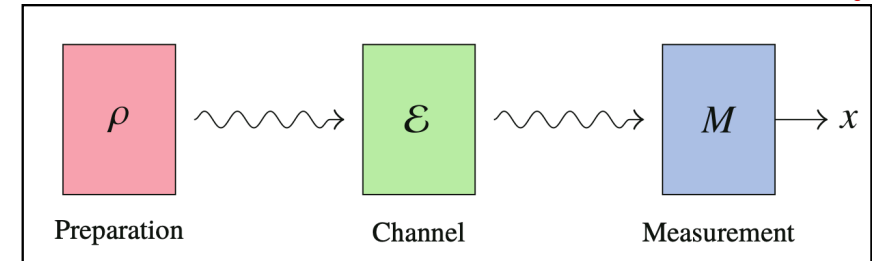
- Recall: Schematic of a quantum experiment
- Quantum channels** describe the dynamics of quantum systems, mapping states from a Hilbert space $\mathcal{H}_A$ to $\mathcal{H}_B$.
- Mathematically:** A quantum channel is an operator:

$$\mathcal{E} : \mathcal{B}(\mathcal{H}_A) \rightarrow \mathcal{B}(\mathcal{H}_B)$$

where $\mathcal{B}(\mathcal{H})$ denotes the set of bounded operators on the Hilbert space $\mathcal{H}$.

- Composite space** $\mathcal{H}_A \otimes \mathcal{H}_E$: Add an ancillary Hilbert space $\mathcal{H}_E$ (environment state ρ_E) to the input space $\mathcal{H}_A$
- Unitary evolution of composite state with output state (using partial trace):

$$\rho' = \mathcal{E}[\rho] = \text{Tr}_E [U (\rho \otimes \rho_E) U^*]$$



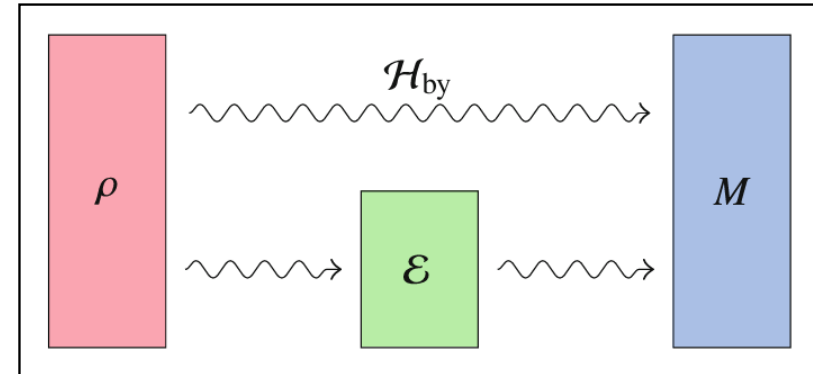
Unitary representation of a noisy quantum channel

Quantum Channels

• **Definition:** A quantum channel is a map that satisfies the following properties:

1. **Linearity:** $\mathcal{E}(\alpha\rho + \beta\sigma) = \alpha\mathcal{E}(\rho) + \beta\mathcal{E}(\sigma)$ for all $\rho, \sigma \in \mathcal{B}(\mathcal{H}_A)$ and $\alpha, \beta \in \mathbb{C}$.
2. **Complete Positivity:** $\mathcal{E}$ is completely positive if for all $n \in \mathbb{N}$, the extended map $\mathcal{E} \otimes \text{id}_n : \mathcal{B}(\mathcal{H}_A \otimes \mathbb{C}^n) \rightarrow \mathcal{B}(\mathcal{H}_B \otimes \mathbb{C}^n)$ maps positive operators to positive operators. This is necessary to ensure positivity when part of a larger entangled system is acted on.
3. **Trace Preservation:** Ensures output is a valid density matrix:

$$\text{Tr}(\mathcal{E}(\rho)) = \text{Tr}(\rho) \quad \text{for all } \rho \in \mathcal{B}(\mathcal{H}_A)$$



- **Summary:** A quantum channel is a linear, completely positive, and trace-preserving map $\mathcal{E} : \mathcal{B}(\mathcal{H}_A) \rightarrow \mathcal{B}(\mathcal{H}_B)$ satisfying the three above properties.
- These conditions ensure that $\mathcal{E}$ maps valid quantum states to valid quantum states in all contexts.



Kraus Representation

- **Theorem (Choi-Kraus):** A map $\mathcal{E}$ is a quantum channel if and only if there exists a set of operators $\{K_j\}_{j=1}^d$ such that

$$\mathcal{E}(\rho) = \sum_{j=1}^d K_j \rho K_j^\dagger, \text{ where}$$

$$K_j : \mathcal{H}_A \rightarrow \mathcal{H}_B \quad \forall j \in \{1, \dots, d\} \text{ with } d \leq \dim(\mathcal{H}_A)\dim(\mathcal{H}_B) \text{ and}$$

$$\sum_{j=1}^d K_j^\dagger K_j = I_A.$$

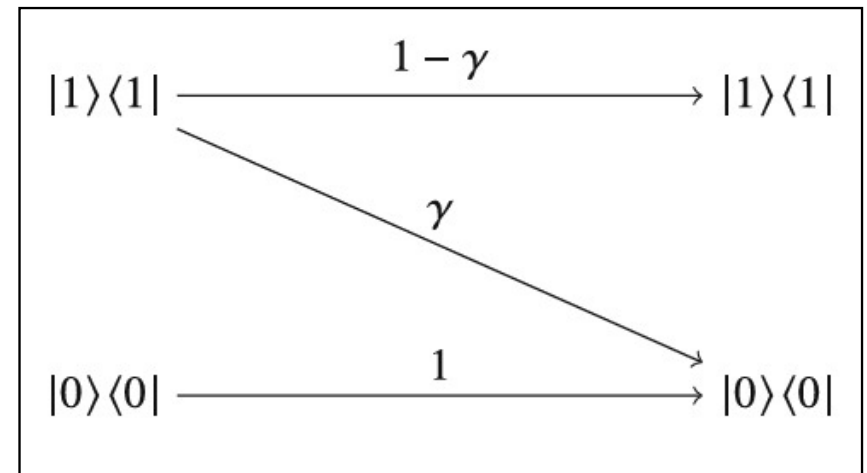
- **Examples:**

- **Unitary Channel:** $\mathcal{E}(\rho) = U\rho U^\dagger \equiv \mathcal{U}(\rho)$ where U is unitary.
 - Only one Kraus operator. It is easy to reverse such a channel.
 - **Reversible:** $(\mathcal{U}^\dagger \circ \mathcal{U})(\rho) = U^\dagger U \rho U^\dagger U = \rho$
- **Isometric Channel:** $\mathcal{E}(\rho) = V\rho V^\dagger$ with $V^\dagger V = I$, $V : \mathcal{H}_A \rightarrow \mathcal{H}_B$.
- **Note:** Kraus decomposition is not unique; different sets of $\{K_j\}$ can describe the same channel.



Quantum Channels Examples

- **Amplitude Damping Channel:** It models energy dissipation (e.g., spontaneous emission) and describes the decay from the excited state $|1\rangle$ to the ground state $|0\rangle$ with probability γ , where $0 \leq \gamma \leq 1$.
 - Kraus Representation:
$$K_1 = \sqrt{\gamma} |0\rangle\langle 1|, \quad (\text{decay from } |1\rangle \rightarrow |0\rangle)$$
$$K_2 = |0\rangle\langle 0| + \sqrt{1-\gamma} |1\rangle\langle 1|, \quad (\text{no decay})$$
 - These satisfy the completeness relation: $K_1^\dagger K_1 + K_2^\dagger K_2 = I$
- **Interpretation:** With probability γ , the system decays to $|0\rangle$, and with probability $1 - \gamma$, it remains in its state.





Quantum Channels Examples

- **Bit flip channel:**
 $\mathcal{E}(\rho) = p\sigma_x\rho\sigma_x + (1-p)\rho$.
- Kraus representation:
 $d_E = 2, K_1 = \sqrt{1-p}I_2, K_2 = \sqrt{p}\sigma_x$

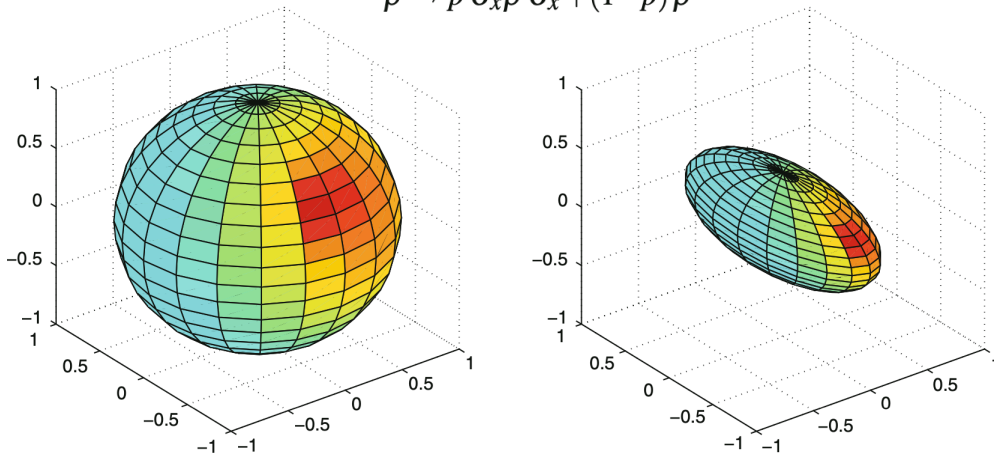
- **Depolarising channel:**
 $\mathcal{E}(\rho) = (1-p)\rho + p(1/d)I_{\mathcal{H}}$.

- Kraus representation can be obtained for finite-dimensional Hilber space.

- For qubit space: $\mathcal{H} = \mathcal{Q} = \mathbb{C}^2$:

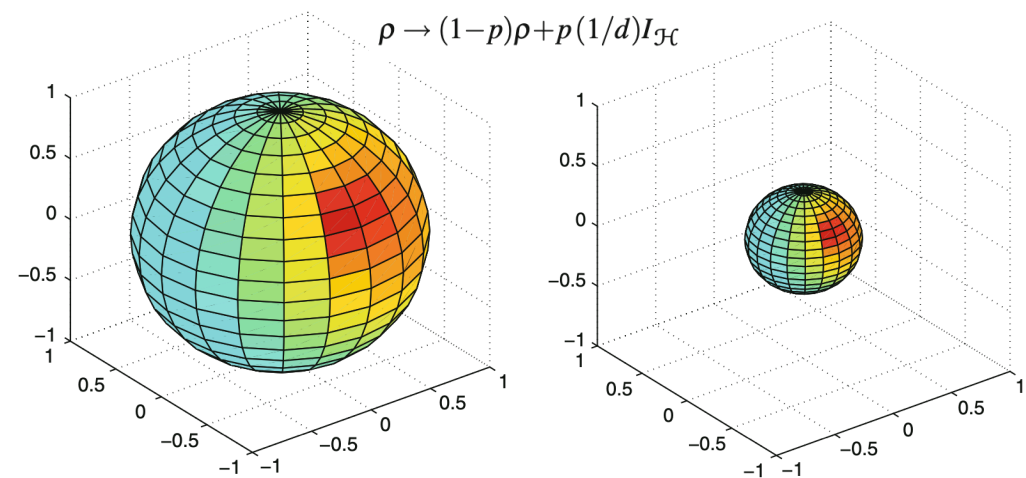
$$I_{\mathcal{H}} = I_2 = \frac{1}{2} \sum_{i=0,x,y,z} \sigma_i \rho \sigma_i^* .$$

$$\rho \rightarrow p\sigma_x\rho\sigma_x + (1-p)\rho$$



Interpretation of the **bit-flip channel** with the *Bloch sphere*.

$$\rho \rightarrow (1-p)\rho + p(1/d)I_{\mathcal{H}}$$



Interpretation of the **Depolarizing Channel** with the *Bloch sphere*.

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Different Notions of Capacity for Quantum Channels

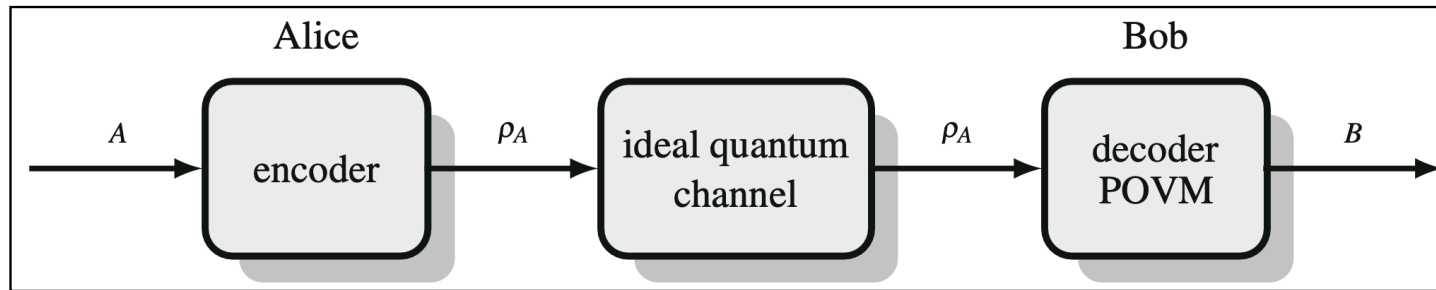


- **Classical Capacity** $C(\mathcal{E})$: Best rate at which classical information (bits) can be transmitted over the channel.
- **Entanglement-Assisted Classical Capacity** $C_E(\mathcal{E})$: Best rate at which classical information (bits) can be transmitted over the channel when the sender and receiver share an arbitrary number of quantum states (ebits).
- **Private Classical Capacity** $P(\mathcal{E})$: Best rate for sending classical information to achieve high fidelity between sender and receiver without leaking information to the environment (Eve). Applicable for secret key capacity of QKD and quantum wiretap channel capacity.
- **Quantum Capacity** $Q(\mathcal{E})$: Best rate for sending quantum information (qubits) over the channel.



Classical Capacity: Accessible Information

- **Classical-Quantum Communications:** Classical information transfer Alice $\rightarrow$ Bob with classical source at Tx.: $A \simeq (\mathcal{A}, p_A)$ and receiver side sees a classical source: $B \simeq (\mathcal{B}, p_{B|A})$



- **Mutual Information:** $I(A; B) = H(A) - H(A | B)$
quantifies how much information A and B have in common
- **Accessible Information:** The maximum mutual information obtained through optimum POVM measurement

$$I_{\text{acc}} := \max_{\text{POVM}} I(A; B)$$